JTMK Project Report:
IoT-Based Smart Lab Energy Monitoring System

Project Title:
Smart Energy Monitoring System for Computer Labs Using IoT

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Introduction:
This project focuses on building an IoT-based energy monitoring system for computer labs under JTMK to help reduce energy waste and promote sustainable practices. The system collects power consumption data and displays it through a real-time web dashboard.

Objectives:

1. To track and log electricity usage in JTMK computer labs.
2. To develop a web dashboard for real-time energy data.
3. To improve energy efficiency awareness among students.
4. To provide data-driven insights for lab management.

Tools and Technologies Used:
• Hardware: ESP32 microcontroller, Current Sensor (ACS712), Wi-Fi module
• Software: Arduino IDE, Firebase Realtime Database, HTML/CSS/JS
• Programming Language: C++ (Arduino), JavaScript

Features:
• Real-time voltage and current monitoring
• Live web dashboard with graph display
• Email alert when usage exceeds thresholds
• Daily energy usage logs for each lab

Implementation Process:

1. Connected current sensors to lab power supplies.
2. Programmed ESP32 to transmit data to Firebase.
3. Built web interface for data visualization.
4. Conducted tests to validate accuracy.
5. Calibrated sensors based on lab measurements.

Results:
• Real-time data successfully displayed on web dashboard.

• Usage trends clearly visualized over time.
• Detected abnormal spikes and reported them.

Challenges:
• Network delays during peak hours.
• Sensor calibration inconsistencies.
• Limited Wi-Fi coverage in some labs.

Future Enhancements:
• Add control module to remotely shut down idle devices.
• Use machine learning to predict usage patterns.
• Mobile notifications and offline data backup.

Conclusion:
The smart energy monitoring system empowers JTMK to manage energy use more effectively, promoting sustainability and operational efficiency in academic facilities.

References:

1. Firebase Realtime DB – https://firebase.google.com/docs/database
2. ACS712 Current Sensor Guide – https://lastminuteengineers.com/acs712-current-sensor-arduino-tutorial/
3. ESP32 IoT Projects – https://randomnerdtutorials.com/projects-esp32/